Socio-Ecological Practice Research
https://doi.org/10.1007/s42532-026-00249-0

RESEARCH ARTICLE

Barriers and opportunities to broader participation in transformative model building for sustainable infrastructure in the Southeastern U.S.

Amanda Fackler[1] · Courtney Cronley[2] · Janey Camp[3] · Jonathan Gilligan[4]

Received: 25 February 2025 / Revised: 16 January 2026 / Accepted: 13 February 2026
© The Author(s) 2026

Abstract

The rapidly changing nature of transportation, telecommunications, and energy infrastructure globally, in part a response to climate change, has led to an increased urgency for innovative solutions in sustainable systems. Moreover, the historical environmental injustices within public infrastructure predicate wider participation in drafting new plans, such that past mistakes are not repeated. The methodology for engaging inclusive and integral partners in sustainable infrastructure model building to advance sustainable, equitable transportation and energy systems remains nascent. Our United States (U.S.)-based exploratory study uses qualitative methods of data collection and analysis, drawing on cross-sectional, semi-structured, in-depth interviews with current academics and practitioners ($N = 14$) from a U.S. National Science Foundation-funded workshop. Our study investigates their experiences and perspectives with broader integral partner collaboration within sustainability planning for Tennessee Valley Authority, a historic public works program that encompasses some of the most economically and socially distressed parts of the U.S. Results converged on three themes that speak to perpetuating infrastructure injustice within systems of energy, telecommunications, and transportation: (1) the intersection of place-based and information inequities, (2) economic inequity, and (3) differing positionalities that drive mistrust and uncertainty; and one theme that spoke to overcoming injustices – the necessity of transformative integral partner engagement versus broader participatory processes. Our findings illustrate the readiness of businesses, organizations, and government entities to engage in a transformative process with community members and integral partners to advance sustainable systems. Moreover, we explore participants' unique experiences and perspectives on the barriers and highlight possible solutions to creating sustainable infrastructure in partnership with integral partners.

Keywords Sustainability · Integral partner engagement · Transformative modeling · Participatory process · Transportation justice

1 Background

Infrastructure in the United States (U.S.) operates through regional system networks of interconnected and interdependent rural and urban communities (Khan and Becker 2020). Traditional approaches and infrastructure systems in the U.S. typically lack long-term sustainability and have contributed to inequities between and within communities (Khan and Becker 2020; National Science Foundation 2021). Individuals and communities experiencing low socioeconomic status in the U.S. and abroad are most likely to bear the burden of the degradation of the natural environment, pollution, human displacement, and other ensuing environmental impacts resulting from unsustainable and traditional infrastructure and energy system development while reaping few of the benefits (Agyeman 2008, pp.751–756; MacDonald 2020,

✉ Amanda Fackler
Amanda.fackler@und.edu

Courtney Cronley
ccronle1@utk.edu

Janey Camp
jvcamp@memphis.edu

Jonathan Gilligan
jonathan.gilligan@vanderbilt.edu

1 University of North Dakota, Grand Forks, USA

2 University of Tennessee, Knoxville, USA

3 University of Memphis, Memphis, USA

4 Vanderbilt University, Nashville, USA

Published online: 25 March 2026

pp.2011–2031). The disparity in the distribution of the risks and benefits of the transportation, telecommunications, and energy infrastructure systems highlights the need for collaborative engagement with communities and integral partners to co-create solutions to the historical and current inequities in these local and global energy systems (Pereira et al. 2017; Sovacool and Dworkin 2015, pp.435–444). However, within the U.S., individuals, community and government agencies, and academia can differ substantially in how they view sustainability, energy systems, telecommunications infrastructure, and the process by which we achieve equitable solutions. Often, researchers' goals and focus are not closely aligned with a community's needs, and there is longstanding recognition that the goals of academia are often disconnected from those of grassroots and advocacy groups (Jenkins et al. 2020). Government agencies struggle to communicate the complex, multifaceted nature of transportation infrastructure, energy systems, telecommunications, and sustainability objectives to their constituents. Even more so, we see disconnects in understanding, desire, and goals among these factions regarding the creation or necessity of sustainable infrastructure. In response, a growing area of research focuses on energy justice and sustainable infrastructure – systems that "ensure economic and financial, social, environmental (including climate resilience), and institutional sustainability over the entire infrastructure life cycle" (United Nations 2022). However, the applicability and efficacy of participatory research and the transformative process in developing sustainable infrastructure, energy, telecommunications systems, and policy pathways with integral partner involvement remain understudied. By the transformative process, we mean the intentional collaboration with community members and integral partners as equal members of the research and change process. The transformative process views the community and its members through a strength-based perspective rather than a deficit perspective that focuses on the structural and energy concerns without regard to the (Mertens 2009, pp.684–695) strengths and expertise of the community and its members. To engage in the transformative process, researchers have developed the transformative modeling conceptual framework, comprising simulation models that support transformative processes by creating simplified versions of reality to understand complex problems and develop novel change pathways (Bidwell and Sovacool 2023, pp.317–320; van Bruggen et al. 2019). However, almost no literature is specific to integral partners or to community perspectives of collaborative justice efforts that utilize transformative modeling or the transformative participatory process.

This paper investigates researchers' and practitioners' perceptions of how integral partners and members of the wider community perceive and experience transportation, sustainable infrastructure, energy, and telecommunications systems, and broad collaboration within the Tennessee Valley Authority (TVA) region in the southeastern U.S. Designing sustainable infrastructure within a changing environment where the risks and benefits are shared equitably is a complex problem that requires a broad range of integral partners (Sovacool and Dworkin 2015, pp.435–444). Therefore, the participation of the public and local communities in addressing, implementing, and sustaining infrastructure changes cannot be understated. Sustainable transportation infrastructure, telecommunications, and energy justice must involve the integration of social and economic equity, (Agyeman 2008, pp.751–756) while evaluating the outcomes of social and environmental vitality (Basiago 1995, pp.109–115). We agree that the participatory process of transformative modeling to develop novel policy pathways to address inequitable infrastructure is a probable solution to the unprecedented challenge of designing and maintaining sustainable infrastructure. Therefore, investigating how researchers and practitioners perceive how integral partners experience the collaborative process, transportation, telecommunications, and energy systems is necessary in the growing literature on sustainability, energy justice, and the transformative participatory processes.

1.1 The Tennessee Valley Authority

This study is part of a larger study seeking to understand the challenges to reliable, sustainable, and equitable access to electricity, transportation, mobility, and broadband telecommunications in the Southeastern United States, with a particular focus on the area served by Tennessee Valley Authority[1] (TVA). Specifically, the larger study sought to better understand the critical challenges different communities face, identify ways future research can help manage these challenges, and design a proposal for a large-scale research network.

The Tennessee Valley Authority (TVA) was established in 1933 by the U.S. Congress as part of President Franklin D. Roosevelt's New Deal to improve the quality of life and economy of the area, bringing electricity to rural areas, constructing a series of dams on the Tennessee River, implementing conservation efforts, and promoting the agricultural development of the region[2]. TVA currently provides electricity to over 10 million people[3], coupled with flood

[1] https://www.tva.com/ Accessed 10 January 2026.

[2] https://www.tva.com/about-tva/our-history Accessed 10 January 2026.

[3] https://www.tva.com/energy/public-power-partnerships Accessed 10 January 2026.

control[4], in-land waterway navigation[5], and land management[6] for the Tennessee River system (TVA 2023). TVA serves 153 local power companies in Tennessee and parts of Alabama, Mississippi, Kentucky, Georgia, North Carolina, and Virginia. In addition, TVA is responsible for cell towers throughout the region, expanding broadband internet, and the fiber initiative, which aims to modernize the region's electric grid to provide all residents with reasonable, non-discriminatory access. Moreover, TVA is helping drive innovation and the adoption of electric vehicles in the region. TVA continues a commitment to environmental stewardship issues as part of its service commitment to effectively use the resources within the Tennessee Valley region – including 293,000 acres [118,572 hectares] of land and 11,000 miles [1609 m] of shoreline – to serve the greater community[7].

TVA also espouses an environmental policy to "produce increasingly clean, reliable, and affordable power, support sustainable economic growth in the Tennessee Valley, and promote proactive environmental sustainability in a balanced and ecologically sound manner" (TVA, 2023). TVA has been considered a model for larger-scale government-led development internationally, and similar inspired projects include integrated river basins in Colombia, India, Iran, the Ivory Coast, and Mexico. As such, the findings from this study may serve as a particularly useful reference for policymakers, researchers, and practitioners seeking to implement the transformative process in areas with similar government-led development projects. A map of the TVA region is provided below (see Fig. 1).

1.2 Integral partners

The term "stakeholder", among other uses, can refer to an individual who marked their claim to land that they were occupying or stealing from Indigenous territories (Simpson 2017). Critiques have brought attention to the potential disrespect of Indigenous people and the perpetuation of the trauma of colonization in using this term to refer to marginalized groups in the participatory process (Gupta 2007, p.316). Therefore, the research team uses the term "integral partners" throughout to reflect our goal of understanding how to engage and center broader, more inclusive partners in the planning and development of sustainable infrastructure. However, this paper does reflect the term "stakeholder" when directly quoting interview participants.

1.3 Transportation justice theory

Transportation justice theory concerns distributional, procedural, and restorative justice in accessibility, opportunity, and the related burdens of transportation infrastructure (Pereira et al. 2017, pp.170–191). Transportation is framed as a civil right and a social good that, when inequitably distributed, leads to social exclusion through a lack of accessibility and opportunity and an increase in transit-related burdens, such as increased pollution, traffic, and accidents, which predominantly affect marginalized populations (Gordon 2015, pp.180–186; Pereira et al. 2017, pp.170–191). The current U.S. transportation infrastructure has been criticized as unreliable, inaccessible, and unaffordable for many, and as lacking foresight and community planning. Access to essential services and resources, such as grocery stores, banks, schools, and job locations, is limited in most communities, leaving those who rely on public transportation without the resources necessary for daily living and improvements in quality of life. A lack of sustainable, safe, reliable, and equitable public transit, primarily in marginalized communities, is representative of a continuation of the historical racist practice of redlining unique to the U.S. and is an example of distributive and transportation injustice (Lamont 2017; Pereira et al. 2017, pp.170–191).

Transportation justice theory prioritizes marginalized groups and emphasizes the importance of participatory processes through community engagement in policy development and transportation infrastructure planning, as part of a focus on procedural justice (Martens et al. 2012). Procedural justice ensures that the decision-making process provides integral partners with access and opportunities to engage in and contribute to transportation infrastructure planning through a participatory process (Lawrence et al. 1997, pp.577–589; Martens et al. 2012). Examples of attempts to engage in the participatory process using a procedural justice framework include holding public hearings and community forums to elicit input from community members, particularly marginalized communities (Lawrence et al. 1997, pp.577–589). However, critiques of the participatory process note that it is time- and resource-intensive and replete with challenges in ensuring that participants from marginalized communities have access to and opportunities to participate in the decision-making process (Blue et al. 2020). Moreover, it can be challenging to ensure that participants accurately represent the community and to mitigate language, ability, and other barriers to access.

Transportation justice theory also seeks to address the historical and systemic inequities inherent in transportation

[4] https://www.tva.com/environment/managing-the-river/flood-management Accessed 10 January 2026.

[5] https://www.tva.com/environment/managing-the-river/how-the-navigation-system-works Accessed 10 January 2026.

[6] https://www.tva.com/environment/environmental-stewardship/clone-land-management Accessed 10 January 2026.

[7] https://www.tva.com/environment/recreation Accessed 10 January 2026.

Fig. 1 Map of Tennessee Valley Authority (TVA) Service Region (https://www.tva.com/ Accessed 10 January 2026.)

infrastructure and repair the harm done within and unto marginalized communities (Martens et al. 2012; Pereira et al. 2017, pp.170–191). This includes advocating for additional funding to improve public transit within historically marginalized and underserved areas. Examples include investing in bike lanes and sidewalks, and expanding public transportation options to provide greater access to employment, education, and essential services and resources (Martens et al. 2012; Pereira et al. 2017, pp.170–191).

Analyzing the burdens of cost and benefit within traditional transportation infrastructure is a necessary step in advancing scholarship, mitigating current injustices, and exploring solutions to the climate crisis and the lack of sustainable infrastructure (Pereira et al. 2017, pp. 170–191). Our paper takes an initial step toward exploring transportation justice and the participatory process from the perspectives and experiences of policymakers, practitioners, and academic partners to understand how they perceive broader community engagement with integral partners.

1.4 Participatory research and the transdisciplinary process

Participatory research is a process that can vary by levels of community and integral partner involvement (Barreteau et al. 2010, p.2; Cornwall and Jewkes 1995, pp.1667–1676;

van Bruggen et al. 2019). At the very least, participatory research is nominal and a means to an end rather than an actual participatory process (Barreteau et al. 2010, p.2; Cornwall and Jewkes 1995, pp.1667–1676). At best, participatory research is a transformative process involving the collaborative efforts of integral partners, the community, and researchers (Barreteau et al. 2010, p.2; Cornwall and Jewkes 1995, pp.1667–1676) by decentralizing power to co-create solutions and understanding (Hooks 2014). The transdisciplinary process (Hadorn et al. 2008) is a participatory process that is action-oriented, challenging the fractured nature of knowledge proliferation and transcending the academic model through affirming contextual knowledge and continuous collaboration with integral partners (Balsiger 2004 pp.407–421; Després et al. 2004, pp.471–486; Horlick-Jones and Sime 2004, pp.441–446; Klein 2004, pp.515–526; Lawrence and Despres 2004, pp.397–405; Pinson 2004, pp. 503–513; Ramadier 2004, pp.423–439). This process requires equitable exchanges and participation with an intersectional (Kimberlé 1989, pp.139–167) and expansive group of integral partners and researchers, a commitment to deliberate attention to underlying power structures (van Bruggen et al. 2019), and the transparency of researchers and systems of power. The process focuses on empowering integral partners as owners of the creative change process and underpins transformative model building, requiring joint action and the co-construction of models through dialogue between researchers and participants, and ensuring participants maintain control of the model once it has been created (van Bruggen et al. 2019). Transformative participatory research supports participants through collaborative approaches in developing policies and plans of action and knowledge sharing, and is an incremental process (van Bruggen et al. 2019). However, there are gaps in understanding how various participants perceive the transformative process and its necessity for creating sustainable infrastructure and energy systems. Conceptually, there is growing acknowledgment and agreement that sustainable infrastructure requires a transformative participatory research process. Still, scholarship lacks representation, understanding, and perspectives of participants, primarily outside the academy. Therefore, our study seeks to address a knowledge gap regarding the perception of integral partner participation as an incremental step toward increasing understanding—and future viability—of the transformative process.

1.5 Transformative models

Transformative models are simulation models used to support the transdisciplinary and participatory process. Creating a simplified version of reality, these simulations support integral partners and researchers in understanding the problem and creating alternative pathways and policy solutions (van Bruggen et al. 2019). Many models regarding the involvement of integral partners have been developed over the past ten years; however, due to fragmentation within the field of transformative modeling, development and coherence are encumbered (van Bruggen et al. 2019). In a systematic review including semi-structured interviews, van Bruggen et al. (2019) identify four approaches to transformative modeling – Group Model Building, Companion Modeling, Challenge-and-Reconstruct Learning, and Generic Collaborative Environmental Model Building. While these models originate from different contexts and do not share theoretical foundations, they employ a transdisciplinary process as part of their participatory research approach (van Bruggen et al. 2019). There is limited research exploring how best to use transformative models with integral partners as part of the participatory research process, though, and to what extent this would be effective in creating policy pathways to sustainable change.

Situated within the larger study seeking to understand the primary sustainability challenges facing the TVA region, this research study explores the following four research questions (1) what attitudes and perceptions do researchers and practitioners express concerning sustainability within the TVA region, (2) what challenges do they perceive related to integral partner collaboration, (3) what current strategies are in place to support collaboration and cooperation between systems and integral partners, and (4) what are their attitudes and perceptions of participatory research within the TVA region?

2 Methods

2.1 Research context – NSF sustainable regional networks workshops

The Sustainability Research Networks (SRN) program[8] is a U.S. National Science Foundation (NSF) program created to understand current sustainability challenges and to support the development of collaborative research in addressing those challenges and overcoming barriers to sustainability. SRNs are transdisciplinary teams that seek to develop novel research and advance collaborations to address the "challenges in sustainability science, engineering, and education" (NSF 2021). The research team received a planning grant under this program to study sustainable transportation,

[8] https://www.nsf.gov/funding/opportunities/srs-rns-sustainable-regional-systems-research-networks/506222/nsf24-533 Accessed 10 January 2026.

energy, and telecommunications infrastructure within the U.S.-based Tennessee Valley Authority (TVA) region[9].

The team utilized cross-sectional, semi-structured, in-depth interviews to answer the four primary research questions. Interviews were based on a 2-day workshop participation funded by the NSF SRN Grant in March 2022. This workshop focused on the challenges to reliable, sustainable, and equitable access to electricity, transportation, mobility, and broadband telecommunications in the Southeastern United States, with a particular focus on the area served by the Tennessee Valley Authority. Invitations to the workshop were disseminated via research team networks and listservs to include integral partners from government, business, community, and civic organizations within the TVA region. Individuals were required to register in advance. Forty-two individuals registered for the workshop. Table 1 summarizes the demographics of the workshop participants.

Table 1 Sociodemographic characteristics of participants

Sample characteristics	*N* (%)
Gender	
Female	7 (50)
Male	6 (42.9)
Prefer not to say	1 (7.1)
Race/Ethnicity	
Asian American	2 (14.3)
Black/African American	1 (7.1)
Hispanic/Latino/a/x	0 (0)
Native HI/Pacific Islander	1 (7.1)
White	9 (64.3)
Prefer not to say	1 (7.1)
Highest level of education	
Associate's degree	1 (7.1)
Bachelor's degree	3 (21.4)
Master's degree	5 (35.7)
PhD, MD, JD, or other doctoral degree	4 (28.6)
Current employment	
Academic	5 (35.7)
Public organization/Department	4 (28.6)
Other	4 (28.6)
Type of work	
Engineer	2 (14.3)
Planner	1 (7.1)
Researcher	5 (35.7)
Social scientist	2 (14.3)
Other	3 (21.4)
Years of experience, Mean ± SD	10.67 ± 10.688
Age, Mean ± SD	23.85 ± 12.818

[9] https://www.tva.com/about-tva Accessed 10 January 2026.

2.2 Sample recruitment and participation

All NSF workshop registrants were invited via email to participate in an in-depth one-on-one interview. Workshop attendees included researchers, government officials, business owners, community members, and persons from civic organizations within the TVA region. While workshop participants represent a diverse set of positionalities, their perspectives and experiences were given equal weight to ensure equal representation of ideas and to avoid bias. Initial invitations were sent two weeks after the workshop, followed by two follow-up emails. If interested, the research team requested that individuals respond directly to the email invitation, and the first author followed up to schedule the interview. Of the 56 attendees, 14 responded to the interview request and subsequently completed the interview.

Analysis indicated that we achieved data saturation through increasingly redundant information emerging with each subsequent interview (Braun and Clarke 2021, pp.201–216; Low 2019, pp.131–139; Saunders et al. 2018, pp.1893–1907). [See below for an explanation of the analytic method.] The final sample comprised a fairly diverse group of participants and was representative of the workshop attendees, although there was a slight over-representation of women among the interviewees (50%, $n=7$). The majority self-described as White (64.3%, $n=9$), nearly a fifth were Asian American (14.3%, $n=2$), and the median age was 23.85 ($SD=12.818$). There was a fairly even split between individuals working within academic (35.7%, $n=5$) versus public (28.6%, $n=4$) and other (28.6%, $n=4$) types of institutions, which was consistent with workshop attendees.

2.3 Ethical protocol

The University of Tennessee's IRB approved the study. Participation required digital informed consent. Informed consent was obtained from all individual participants included in the study. Workshop attendees were informed that they were free to withdraw from the study at any time and could choose not to answer any questions on the demographic survey and during the semi-structured interviews. Participants consented to the recording and subsequent transcribing of interviews. Interviews were conducted over Zoom and uploaded to the PI's password-protected computer. Interviews were anonymized and stored on a password-protected computer.

2.4 Data collection

Interviews were conducted on Zoom with a single individual at a time and lasted approximately 40 to 60 min. Interviews began with an online informed consent and demographic

survey hosted in Qualtrics. The interviewer provided the survey link in the Zoom chat. Participants were first asked to read the informed consent, give digital consent to proceed, and then complete the eight-item demographic survey. Once interviewees completed the informed consent and demographic survey, the interviewer began recording on Zoom.

All participants were asked the same seven questions; one served as an introduction, and two sought evaluative feedback about the workshop. Four questions were posed to answer the primary research questions of this study and included: *(1) In your opinion, what are the critical sustainability issues relating to telecommunication, energy, and transportation in connecting the rural and urban south?; (2) What are the challenges of stakeholders working together with researchers?; (3) What are the effective strategies of working together among community organizations, utilities, policymakers, and researchers?; and (4) What do you think Tennessee Valley Authority can do to help to improve sustainability in energy, telecommunication, and transportation infrastructure in rural areas?*

2.5 Data analysis

Interviews were recorded and transcribed, and data were saved within Zoom. Zoom audio files were re-transcribed through a professional transcription service for clarity of analysis purposes. Data were then analyzed using an incremental, descriptive content analysis from open coding to first-order themes and second-order themes to move from abstract data to more concrete interpretation (Lindgren et al. 2020). The first and second authors read and re-read each transcript and identified potential themes. Each author reviewed these initial codes, informed by the four primary research questions, and produced overarching themes and sub-themes. In the next stage, each author identified salient quotes to the overarching themes. Lastly, each author reviewed the themes to define and name them. Rigor methods included paper trails, triangulated coding, and member checking (Cresswell 1998).

3 Results

3.1 U.S. barriers to broader collaboration – In/Equities

Across interviews, a central idea emerged: broader perspectives and possible solutions for sustainability planning in transportation, telecommunications, and energy justice *hinge on equity*. Specifically, the themes of *place-based and information inequities*, *economic inequities*, and *differing positionalities* that drive continued participatory inequities in planning for a more equitable and sustainable future infrastructure.

Throughout interviews, participants consistently reported a "lack of equity" as a significant impediment to sustainability and collaboration in telecommunications, energy, transportation, and connecting the rural and urban South within the TVA region. For example, in discussing access to infrastructure within the TVA region, one participant noted; "Equity. It's really about access, right?… the physical access and the economic access are probably the biggest things that I think about." Another participant noted that considerations of equity need to go beyond physical and economic access, stating, "We focused a lot in the workshop on income, and I think we need to do more work on like *race and gender*."

Participants identified equitable access to transportation and telecommunications infrastructure as critical to building sustainable infrastructure within the TVA region, and they observed that such access differs markedly across place and space. They discussed the inequities between the rural and urban South, and between marginalized communities and communities of privilege. These place-based inequities were highlighted as major concerns and impediments to wider participation with integral partners in sustainability planning. These inequities of daily life were seen as barriers to engagement, as participants from marginalized communities were viewed as more likely to lack the economic and physical access to participate. For example, one participant highlighted the disparity between income and transportation burden experienced by individuals in rural areas, stating, "You know, folks in rural areas have *a lower income*, but they have a *higher transportation burden* in comparison to their income." Another participant, discussing the lack of sustainable telecommunications within the region and the disparities between rural and urban areas, stated, "Rolling out municipal high-speed fiber to everybody. That could solve that equation and get *everybody on the same playing field* as far as rural and urban life."

We learned from participants that the very infrastructural inequities the participatory process seeks to address —such as equitable access to reliable transportation, access to opportunity, high-speed internet, and emerging technologies —are also perceived as barriers preventing integral partners from marginalized communities from participating. And, these barriers were understood to differ by place, specifically, whether integral partners live in rural or urban areas. These place-based inequities were seen as an initial and ongoing barrier to integral partner participation in the transformative process and were discussed as part of the primary themes throughout.

3.2 Transportation, internet access, and telecommunication backbones

The topography and land use of the TVA region prioritizes travel by personal vehicle, and there are disincentives to using public transit, e.g., longer commute times, first- and last-mile inconvenience, and limited destinations. And in rural areas, specifically, the only public transit is usually paratransit. As one participant noted, "We are in an area that is really set up to prioritize and encourage single-passenger vehicle trips." From a similar perspective, discussing the lack of public transit options, another participant mentioned, "I mean, how many people take, really take, a Greyhound, right? So intercity conductivity and things are important."

Participants shared that within the TVA region, individuals with lower incomes disproportionately use public transportation, and there is public sentiment that, because they are lower-income, they are expected to spend more time on transportation than those who own their own vehicle. As one participant stated:

> If you're low-income, there's no other way than to take a bus…You have to spend a lot of time and then you don't make money, so there's also the time cost of money. If you're poor, I mean, you're *expected to waste more time on transportation*.

And not only is there a perception that if you are using public transportation, you are expected to "waste time," but as another participant added, there is a public "…perception problem of 'buses are for poor people' and people like having that mental block that somehow buses are for only poor people" Along with the reality that in the TVA region, "The bus system isn't very responsive or efficient for other people's needs" these perceptions serves as barriers for individuals with higher incomes and personal vehicles in understanding or seeking sustainable solutions to these very real transportation inequities. Moreover, as participants suggested, this perception is an explicit example of the historical and systemic injustices faced by marginalized communities in public transportation. Thus, the idea of electrifying public transit, embedded in TVA's sustainability goals, given its apparent underperformance and inefficiency, seems out of sync with the priorities and needs of integral partners—the first priority being a more equitable and efficient public transit system.

In discussing TVA's goal to promote electric transit, participants specifically discussed the potential barriers to integral partner engagement and adoption of electric vehicles as a sustainable solution. As one participant noted, if integral partners are interested in adopting electric vehicles, there remains a "lack of supply," preventing those who can afford them from purchasing them. This same participant added that, even with an adequate supply of electric vehicles, they would remain unaffordable for lower-income individuals, stating, "They just don't have that capital to outright pay for an electric car, even a used electric car…. You know, the vehicles they get are very fuel inefficient because they've just been passed down." And referencing place-based inequities, one participant noted that even with the resources to purchase an electric vehicle, rural populations would be unable to take advantage of the cost-saving benefits due to a lack of "*telecommunication backbones*" in the rural South, stating, "If you don't have those telecommunication backbones available for people in rural areas, then they can't take advantage of some of these cost-saving metrics for owning an electric vehicle."

Emerging as a critical issue and barrier to transportation were discussions about the need for a secure, reliable internet connection for all communities. An unreliable internet connection and lack of telecommunication backbones were identified as impediments to sustainability planning and to the broad participation of integral partners. As another participant noted, discussing a recent power outage at their home and highlighting the intersections of income and place-based inequities, "We do have an electric backup, but not everyone has that; *not everyone is able to afford that.*" Participants elaborated that the lack of and inequitable distribution of a sustainable internet connection further perpetuates these place-based and economic inequities. As one participant stated, "Ultimately, we need to have a secure and sustainable broadband connection…for *basic health and safety.*" Other participants noted that the lack of broadband internet is especially problematic in rural areas, with impacts to present and future educational opportunities, "[The need for] having adequate high-speed internet, as far as education," and:

> I do think that rural broadband is extremely important… I know that it's an issue around *equity* and like people being able to have good jobs and people being able to have like online school, and that was a big issue in some areas where we didn't have computers or internet or fast enough internet.

Along these same lines, another participant stated, "And I do know, personally know, some people that the best internet they can get is satellite internet, which is still really slow."

Participants therefore discussed how the incentives for integral partners to participate in the transformative process are low, especially for individuals without the economic opportunity to purchase newer technologies, such as an electric vehicle, or afford access to the telecommunications system to benefit from its use. Perceptions of public transit

and its efficiency were seen as barriers to participation in the transformative process for integral partners in more urban, marginalized communities, whereas access to telecommunications, such as a reliable internet connection, was perceived as a barrier for integral partners in more rural areas. Economic accessibility and access to a reliable telecommunications system were seen as barriers to the participatory process and to transformative modeling with marginalized communities, especially as they relate to building sustainable solutions for public transit, adopting electric vehicles, and telecommunications within the TVA region.

3.3 Economic In/equity

Issues related to financing sustainable infrastructure emerged as a barrier to broad and equitable integral partner engagement, and not just inadequate or limited federal funding, but also the use of funding and its impact on sustainable infrastructure. For example, one participant discussed how rural areas are more likely to receive less funding than their city counterparts, further perpetuating inequity, stating, "Connection of rural and urban in the urban south? With that one, I think it's funding." Without the necessary funding for infrastructure improvements in the rural South, participants felt there was no incentive from a policy or community member standpoint to engage in the participatory process. And yet, as one participant noted, there are concerns about the existing gaps in telecommunications and transportation coverage, and more specifically, the need for a participatory process to determine where to allocate federal funding based on vulnerability and need.

> "We're going to invest federal transportation dollars in rural areas; we're trying to figure out where to get the most bang for our buck. Where do you have the most vulnerable populations?"

Further compounding efforts to engage more diverse integral partners in shifts towards more sustainable infrastructure, participants discussed how place-based inequities have been historically perpetuated rather than mitigated by emerging sustainable transportation and energy technologies, with one participant noting, "I think a really big issue is green gentrification." And as another participant noted, "I don't think many people think about how renewable energy or clean energy or anything like that can like worsen social inequality." For example, rural and urban underserved communities face connectivity challenges related to a lack of reliable high-speed internet. Yet, some of these communities are home to energy plants and internet towers, providing access to other areas while access remains unavailable or limited within their community. As one participant noted, "One woman was telling me she's got a huge internet tower in her backyard, but she actually can't get internet." Along those same lines, another participant noted that "Rural areas may not benefit from resources to which they contribute because those resources are being funneled to urban areas." While another participant discussed how "communities of people who are majority black or majority working class are bearing the brunt of you know, the electrified world."

These disparities in resource allocation and the inequitable distribution of the risks and benefits associated with emerging infrastructure were perceived as barriers to broader integral partner participation and the development of sustainable solutions.

3.4 Differing positionalities, mistrust, and uncertainty

Emerging as a key theme, *differing intersectionality and positionality* were viewed as potential barriers to integral partner collaboration. Participants shared how differing expectations among integral partners, academia, and government officials shape communication. These differences can sow mistrust or uncertainty among different parties and undermine community-wide sustainability initiatives. As an example, one participant noted that publications are less valued outside academia and that "stakeholders don't care whether the research was published or not." The historical focus on papers has eroded confidence among community partners who are more interested in tangible benefits for their communities' needs. Moreover, as another participant stated, "They [community members] don't necessarily understand research or what a quasi-experimental design is. Certainly not what a randomized control trial is, and they don't necessarily care about those things." And not only is there a disconnect in values, but as another participant, working in academia, discussed, "in my own experience, I will come in and use jargon that is not understood." The use of researchers' jargon, a lack of shared language, and an emphasis on data and papers were all considered as hindrances to collaboration with community agencies and integral partners.

As such, participants indicated that in their experience, community agencies and integral partners are concerned about being taken advantage of by researchers. Participants discussed how agencies feel that researchers enter communities to advance their own agendas, with little regard for community members' expertise or the needs of community agencies, and as one participant noted, "[Researchers] often seek the assistance of stakeholders and community agencies with little to no return on stakeholder investment." Community agencies and integral partners have priorities that differ from those of researchers and are often working

to benefit the community with limited funding, as one participant noted: "Stakeholders are often really busy doing things researchers can't imagine – fighting fires and trying to raise money." Unfortunately, as another participant, who previously worked with numerous non-profits, indicated, researchers often "assume they know more than the community," which can further undermine trust and collaboration.

In addition to experiences of mistrust and uncertainty regarding shared goals, there is a sentiment that researchers do not value community members for their experiential knowledge. As one participant, discussing workshops and community meetings, stated, "I don't think [researchers] often see [community members] as a valuable source of information, often like a lot of the boards that were presenting at the workshop…they don't allow community members into their meetings and it's like, well, why not?" And, as another participant noted, there are "[A]ll kinds of power imbalances and internalized superiority even of researchers in general to like lay, people, you know, or to even to non-profit leaders." Moreover, racial and place-based biases were also discussed as barriers to collaboration between researchers and stakeholders. For example, one participant noted they had experienced bias due to intersecting identities of race and place, stating, "[M]ainly racial bias, but also there's a bias against you from the south."

The theme of differing positionalities was further developed as participants discussed the challenge of "getting all of the integral partners at the table [including] disadvantaged, vulnerable communities" to tap into the "lived experience" of affected community members. Participants discussed that without "equitable access to venues" and "equitable internet access," it is challenging to gather an accurate representative sample of the community and "identify those trusted leaders - informal and formal." Participants noted the importance of meeting with local government and elected officials to align with their agendas while communicating early and often with the local community – acknowledging how this can be a challenge. As one participant noted, "And one of the challenges that we have is getting them all on the same page to agree to here's how we want to prioritize projects that come before us for federal funding." Even more so, participants noted the challenge of discussing infrastructure proposals without a shared language, stating, "And those conversations are tough to translate down to that Community level because they are very new ones [conversations about sustainable infrastructure and new technology], and there's a lot of details."

The overarching theme of barriers to collaboration and sustainable infrastructure was a lack of equity—place, knowledge, economics, and position. Participants discussed how sustainable infrastructure is only possible to the extent that we can challenge systemic social and economic inequity. Collaboration among integral partners, researchers, agencies, and local government officials is often maligned by the imbalance of power resulting from positionalities of inequity. The disparities and oppression due to the varied intersections of class and place are the primary barriers to collaboration toward sustainable infrastructure.

3.5 Transformative modeling – participatory processes to overcome inequity

Throughout the interviews, participants consistently moved from discussing barriers created through inequity and positionality to potential solutions and experiences of success. Five primary themes emerged as examples of the transformative participatory process: *creating a collaborative language of sustainability*, *the importance of messaging*, *a collaborative process*, *overcoming inequities*, *and continued engagement.* Participants discussed the need to recognize different positionalities, invest in communication, and create a shared language to generate equity in collaboratively designing solutions. As many participants noted: "*a shared language would be helpful.*" Based on the *Typology of Approaches to Modeling with Stakeholders* (van Bruggen et al. 2019, p.8), illustrating the various typologies of

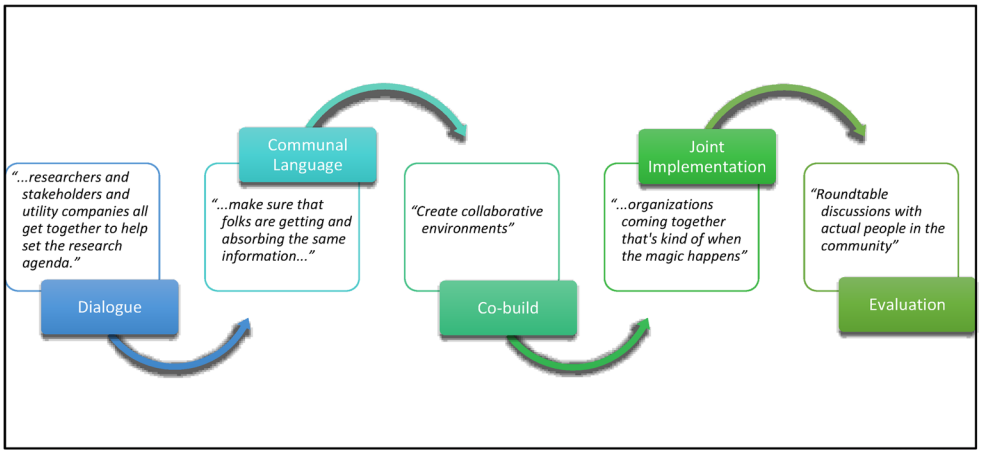

Fig. 2 Participant Perspectives of the Transformative Process (based on the table from *Modeling with integral partners for transformative change (*van Bruggen et al. 2019, p.8*)*)

approaches to modeling with stakeholders, Fig. 2 highlights the main themes from participants through direct quotes, underscoring the transformative participatory process consistent with the transformative modeling approach addressed therein.

3.5.1 Creating a shared language of sustainability

Participants discussed the importance of clear communication among academics, government officials, community agencies, and integral partners. They called for the creation of a *collaborative language of sustainability* that would enable participants in the participatory process to engage in effective dialogue. As one participant stated, "It's just knowing language is so important, just knowing how to connect with people." Participants wanted to ensure that all integral partners had equal opportunity to understand and share their perspectives without having to navigate "researchers' jargon." In the words of one participant, "You know, meeting people where they're at, no matter where that is, you know, we always talk about meeting people where they're at if they're like, you know, not having the same education or whatever." Participants shared that from their experience, engaging in open communication with integral partners increased the success of implementing sustainable infrastructure.

3.5.2 The importance of messaging

Participants also highlighted the importance of *effective messaging*, specifically to ensure that all participants had access to the necessary information. For example, one participant stated the importance of "educating the public around what public service commissions do and how important they are." While another participant noted that "having a facilitation and a facilitator that can make sure that folks are getting and absorbing the same information is going to be important."

3.5.3 A collaborative process

Participants emphasized the need for early, frequent communication with community leaders and integral partners throughout the *collaborative process.* More specifically, participants stated the importance of having "…open and early conversations with people on a regular basis," and holding "meetings where the researchers and stakeholders and utility companies all get together to help set the research agenda." As another participant noted, meetings should be conducted, "at every point of the process from the early ideas to say, ' We've identified some challenges, here are some potential solutions, and that needs to be happening sometimes years before you get to an actual project proposal." Moreover, participants focused on the need to ensure the process originated at the community level, stating, "for the communities to be the number one thing is a needs assessment from the ground up," and encouraged a process where researches and government agencies "[make] the effort to communicate with them and setting up a time to talk and say hey…here's what we would like to see and what your plans are, how can we get together and work with one another." Furthermore, participants added, "And then to listen to community members listen to what they are concerned about in their community." And, as one participant noted, "When you have these organizations coming together, that's kind of when the magic happens."

3.5.4 Overcoming inequities

Participants discussed possible solutions to *overcome inequities* and mitigate barriers to building collaboration with community members and integral partners—specifically, ensuring that integral partners and community members have adequate internet and transportation to local or virtual meetings and are compensated for their time and expertise as appropriate. For example, one participant stated that it is essential to ensure "equitable access to the venues, to the information, to the conversations." At the same time, another noted the importance of "[having] the time and space and education and access to the Internet and everything else to participate in these kinds of conversations." This might require forward thinking, as one participant suggested, "To give folks an equal footing, it may be providing computers, it may be providing internet access, and certainly another key part of it is providing information." It might also look like "sending buses more regularly for 10-minute headways and also sending them further out to rural communities." Participants also discussed overcoming inequities through resource sharing, more specifically, one participant stated, "Sharing resources, including data, monetary, financial, all of those things, as well as authentic engagement between all of those parties." And, as another participant noted, "Paying people for their lived experience, you know, to share their lived experience, just like you would pay other consultants to come to the table and provide their expertise."

3.5.5 Continued Engagement

Lastly, participants discussed the need for *continued engagement* in the community and the collaborative process. Participants felt there should be "opportunities for feedback and engagement that help you build the trust that you need to then move forward," and "roundtable discussions with actual people in the community," so that "they

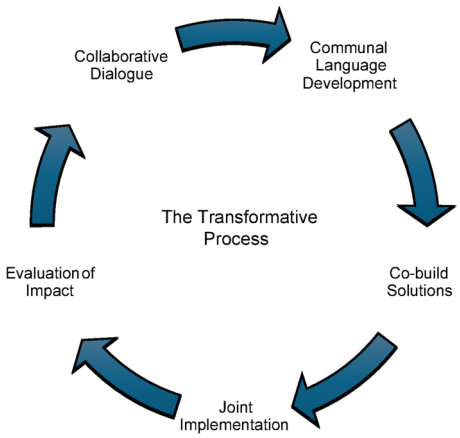

Fig. 3 The Transformative Process (based on table from *Modeling with integral partners for transformative change (*van Bruggen et al. 2019, p.8*)*)

can communicate on a frequent basis what they're going through that so that people feel that it's a transparent process."

Based on the primary themes identified as keys to the participatory process, we submit an alternative method to the traditional participatory process model, rooted in modeling for transformative change (van Bruggen et al. 2019), namely the Transformative Process. Figure 3 illustrates the proposed approach, highlighting the need for a cyclical, ongoing, inclusive, and iterative process.

4 Discussion

The inherent and historical inequities embedded in the current telecommunications, energy, and transportation infrastructure in the United States also pose a central challenge to creating sustainable and equitable change through collaborative partnerships and a transformative process. It should be noted that Title VI of the Civil Rights Act in the United States bans any organization or entity receiving federal funds from denying services or goods or discriminating against any individual based on race, color, or national origin, and TVA is directly mandated to comply with Title VI. As such, the persistent discussion regarding equity in access to public infrastructure suggests a slippage between federal policy and on-the-ground implementation, and the importance of research such as this. Within the U.S., the entrenched inequities and racially motivated redlining, or discriminatory denial of services to residents of particular socio-cultural demographics, underscore the critical importance of broader integral partner engagement in policy and infrastructure planning. This type of transformative process would ensure that groups in positions of power do not dictate the terms of sustainability and, instead, that societies can broadly distribute the benefits of future sustainability plans (National Archives and Records Administration 1980).

Our findings highlight awareness and a desire among academics, business owners, and government officials to achieve a more transformative planning process in the TVA region. However, participants continually cited existing infrastructure barriers —some unique to the TVA region and others characteristic of the U.S. as a whole —that impede future equitable planning (National Archives and Records Administration 1980). Place-based and information inequities within the existing systems are significant practical impediments to collaboration with integral partners in the TVA region. Individuals who are digitally redlined and most in need of broadband, specifically those in the rural south, are less able to attend Zoom meetings to discuss plans to expand Internet technologies (Prieger 2013, pp.483–502; Reddick et al. 2020; Singh et al. 2006) or voice opposition to or support of proposed infrastructure. People living in rural communities or without cars are most in need of more transportation options (Gray 2004, pp. 172–181) but are less able to attend in-person meetings on the topic. Participants discussed the need to expand policies increasing access to transportation options as an incremental step in the transformative process. Individuals marginalized by infrastructure systems may also face knowledge and information barriers, leading them to be less aware of emerging technologies and their potential benefits (Du et al. 2020). Participants discussed how community agencies and integral partners face different daily challenges and have fewer incentives to engage in the transformative process due to these knowledge and information barriers. These barriers seem obvious, and yet, the pervasive and embedded nature of inequities in infrastructure may make persons who regularly experience the benefits of infrastructure, such as subject matter experts in the knowledge economy, industry integral partners, and public officials, blind to these barriers in practice, even when they know that they exist.

In addition to place-based information and economic inequities, conflict arising from differing positionalities poses distinct challenges for broader engagement with integral partners. Sustainable solutions require inclusive research endeavors, transparency in motivation, and expectations from academics to communities and integral partners (Lee 2020, pp.1540–1551). Moreover, participants highlighted that it requires acknowledgment of the long-term, arduous work of community, integral partner, and activist labor. The impact of language barriers between researchers and integral partners was an important finding from our interviews, and participants suggested that this should be addressed by creating a shared or communal language regarding sustainability challenges – a collaborative language of sustainability. Some participants also noted success with an early,

iterative engagement process that integrates integral partner expertise into all stages of the research process, offering a blueprint for the transformative process. The application of the proposed transformative process necessitates further participatory research with integral partners to create a shared language grounded in their unique experiences and challenges within an inequitable infrastructural system.

While not novel, forming and engaging with community advisory boards was mentioned by participants as one way to build trust and investment in proposed sustainability solutions and the transformative process. Specifically, participants highlighted the need for community advisory boards that not only include members from marginalized communities affected by the proposed infrastructure changes but also treat them as equal partners in the decision-making process. The challenge for researchers with community advisory boards (CAB) is how to operationalize and work with truly transformative, critical boards. For example, the TVA's Regional Energy Resource Council (RERC) is a community advisory board established in 1995 to provide input on the TVA's energy resource activities (TVA 2024). The council consists of twenty members from academia, environmental organizations, industry, and community interests. While the RERC does provide input into the TVA's energy resource planning activities, there are critiques that the RERC lacks diversity, is dominated by industry representatives, and lacks the influence to alter the TVA's activities. Furthermore, examples from the literature tend to illustrate participatory rather than advisory boards – they serve the function of checking in with the community or understanding policy or regulatory landscapes (van Bruggen et al. 2019). There have been few published examples of CABs embedded in large-scale engineering and planning projects that have pushed back against the plans, much less rerouted the course. More often, such critical engagement comes from the outside. Thus, the extent to which the tension of alternative perspectives may fail to be captured in models and scenario plans.

Additionally, rather than developing outcome measures a priori, participants suggested that researchers and practitioners work with community members to establish meaningful outcome metrics and community-relevant impacts. Such metrics might aid in making the effects of any proposed solutions more transparent – helping to build trust – and in exposing whether solutions are providing value to communities and whether they have potential unintended negative consequences that could be mitigated.

Furthermore, engaging with integral partners to apply a critical focus on equity is essential in creating sustainable solutions to public infrastructure (Jasanoff 2018, p.13). Our participants spoke to how many solutions were reintroducing gentrification and privileging those with greater access to sustainable infrastructure while further disadvantaging those with fewer existing resources. Equity scans may shed light on where additional or differential investments are needed to produce equal access to sustainable technologies (Sovacool et al. 2022, pp.326–337).

The transformative process called for in this paper perhaps also represents an iterative improvement upon the action research (Greenwood and Levin 2006) model in sustainability scholarship. While this model seeks to derive co-designed locally-relevant models of sustainability among academics, civic partners, and communities, it has proven to have inconsistent outcomes (Keahey 2021, pp.291–306). Transformation may involve more than broad participation but also a recognition a priori of the existing inequities and how to strategically overcome those inequities to ensure authentic and sustained participation for those most vulnerable to inequities in public transport and infrastructure. There is a need to critically apply the transformative process outlined in this study to explore further the experiences of integral partners outside academia, industry, and government who are most affected by proposed infrastructure changes.

5 Conclusions and limitations

Consistent with the growing field of participatory research, our findings support the need for a transformative process that involves the intersection of academia and community members as co-owners in creating sustainable solutions to public infrastructure – transformative modeling solutions (van Bruggen et al. 2019). Our findings illustrate the readiness of community organizations, businesses, and academics in the TVA region to participate in the transformative process. Furthermore, our findings add to the growing literature on the transformative process by drawing on the unique perspectives and experiences of our participants regarding the barriers and possible solutions to creating sustainable infrastructure. It is important to acknowledge that this study is limited by a smaller sample size ($n = 14$) of predominantly academic and professional individuals. Our sample lacks representation from community members and those most directly affected by infrastructure inequities, further underscoring the existing disconnect and the need to engage in the transformative process. Further research should seek to identify and understand the lived experiences of community members and those most affected by infrastructure and transport inequities. This requires mitigating barriers to access by creating time and space for all integral partners to gather, creating a communal language, and providing ample opportunities for integral partner feedback and engagement.

Our study's findings must be interpreted within the unique context of the U.S. TVA region, which includes large rural

and economically distressed areas. And yet, our participants spoke to themes that resonate with the existing literature. It may be, though, that in more urban communities, transformative model-building and the transformative process may look different. In addition, our data were collected as societies reopened post-COVID, and issues of public trust and community engagement were severely dampened during the pandemic. Thus, our findings may reflect this historical bias. Doubtlessly, the nature of communication processes and the types of sustainability challenges may vary over time and across contexts. However, the lack of diverse, sustained, and integral partner-embedded participation will likely persist across land use and communities.

Overall, sustainable solutions to energy and transportation infrastructure require that researchers and policymakers operate under the principle of equity to ensure a fair distribution of future benefits and shared risk. Doing so is challenging given the current inequities built into our public infrastructure, which limit the potential for transformative solutions in the future. Broadening participation in the process of modeling may offer opportunities for transformative and sustainable solutions. Still, it requires intentional and sustained outreach, co-ownership of ideas, plans, and outcomes, and a willingness to hear and respond to critical feedback. With CABs and improved community engagement, some consequences cannot be anticipated, but ideally, all parties are aware and can take ownership of the outcomes, whatever they may be. As we move toward realizing and adopting transformative modeling more broadly in sustainability planning and engineering, we may test how community-engaged, critical local processes do, in fact, contribute to greater equity in modeling sustainable infrastructure and, ultimately, building resilient infrastructure systems that serve all communities and individuals.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s42532-026-00249-0.

Acknowledgements The authors would like to thank all the integral partners who engaged in the workshops and interviewees who took the time to participate in the interviews. We would also like to thank other members of the research team, Dr. Chien-fei Chen (Co-PI), Dr. Charles Sim (Co-PI), and Dr. Melody Huslage (research assistant).

Author Contributions Conceptualization, C.C. and A.F. and J.C. and J.G.; methodology, A.F. and C.C.; validation, A.F., C.C., J.C. and J.G.; formal analysis, A.F. and C.C.; investigation, A.F. and C.C.; resources, C.C. and J.C. and J.G.; data curation, A.F. and C.C.; writing—original draft preparation, A.F. and C.C.; writing—review and editing, A.F., C.C., J.C. and J.G.; visualization, A.F. and C.C.; supervision, C.C., J.C.; project administration, J.G., J.C., & C.C.; funding acquisition, J.G., C.C., and J.C. All authors have read and agreed to the published version of the manuscript.

Funding The research leading to these results received funding from the UNITED STATES NATIONAL SCIENCE FOUNDATION, under grant agreement No. 2115392.

Data availability Data available on request due to restrictions e.g., privacy and ethical. The data presented in this study are available on request to the second author. The data are not publicly available due to privacy and confidentiality of interviewed participants.

Declarations

Consent to publish Participants provided informed consent regarding publishing their data.

Human ethics and consent to participate Informed consent was obtained from all subjects involved in the study.

Institutional review board statement The study was conducted in accordance with the Declaration of Helsinki, and approved by the Institutional Review Board of The University of Tennessee (protocol code UTK IRB-22-06865-XM, March 23, 2022).

Conflicts of interest/competing Interests The authors have no relevant financial or non-financial interests to disclose. The authors have no conflicts of interest to declare that are relevant to the content of this article. All authors certify they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript. The authors have no financial or proprietary interests in any material discussed in this article. The authors declare no conflict of interest. The funders had no role in the design of the study; in the collection, analyses, or interpretation.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Agyeman J (2008) Toward a 'just'sustainability? Continuum 22:751–756. https://doi.org/10.1080/10304310802452487

Balsiger PW (2004) Supradisciplinary research practices: history, objectives and rationale. Futures 36:407–421. https://doi.org/10.1016/j.futures.2003.10.002

Barreteau O, Bots PWG, Daniell KA (2010) A framework for clarifying participation in participatory research to prevent its rejection for the wrong reasons. Ecol Soc 15:1. https://doi.org/10.5751/ES-03186-150201

Basiago AD (1995) Methods of defining 'sustainability'. Sustain Dev 3:109–119. https://doi.org/10.1002/sd.3460030302

Bidwell D, Sovacool BK (2023) Uneasy tensions in energy justice and systems transformation. Nat Energy 8:317–320. https://doi.org/10.1038/s41560-023-01217-8

Blue G, Bronson K, Lajoie-O'Malley A (2020) Beyond participation and distribution: a scoping review to advance a comprehensive justice framework for impact assessment. 1–59. http://hdl.handle.net/1880/112213

Braun V, Clarke V (2021) Questioning data saturation as a useful concept for thematic analysis and sample-size rationales. Qual Res Sport Exerc Health 13:201–216. https://doi.org/10.1080/2159676X.2019.1704846

Cornwall A, Jewkes R (1995) What is participatory research? Soc Sci Med 41:1667–1676. https://doi.org/10.1016/0277-9536(95)00127-S

Cresswell J (1998) Qualitative inquiry and research design choosing among five traditions. Sage, Thousand Oaks, CA

Després C, Brais N, Avellan S (2004) Collaborative planning for retrofitting suburbs: transdisciplinarity and intersubjectivity in action. Futures 36:471–486. https://doi.org/10.1016/j.futures.2003.10.004

Du JT, Xie I, Waycott J (2020) Marginalized communities, emerging technologies, and social innovation in the digital age: introduction to the special issue. Inf Process Manag 57:102235. https://doi.org/10.1016/j.ipm.2020.102235

Gordon C (2015) Varieties of transportation justice: US transportation equity policy and the civil rights movement. Transp Res Rec 2531:180–186. https://doi.org/10.2139/ssrn.2388554

Gray D (2004) Rural transport and social exclusion: developing a rural transport typology. Built Environ 30:172–181. https://doi.org/10.2148/benv.30.2.172.54310

Greenwood DJ, Levin M (2006) Introduction to action research: social research for social change. Sage, Thousand Oaks, CA

Gupta T (2007) Race and racialization: Essential readings. Canadian Scholars', pp 316–324

Hadorn GH, Hoffmann-Riem H, Biber-Klemm S, Grossenbacher-Mansuy W, Joye D, Pohl C, Wiesmann U, Zemp E (2008) Handbook of transdisciplinary research. Springer, Dordrecht, Netherlands

Hooks B (2014) Teaching to transgress. Routledge, New York, NY

Horlick-Jones T, Sime J (2004) Living on the border: knowledge, risk and transdisciplinarity. Futures 36:441–456. https://doi.org/10.1016/j.futures.2003.10.006

Jasanoff S (2018) Just transitions: a humble approach to global energy futures. Energy Res Soc Sci 35:11–14. https://doi.org/10.1016/j.erss.2017.11.025

Jenkins KEH, Stephens JC, Reames TG, Hernández D (2020) Towards impactful energy justice research: transforming the power of academic engagement. Energy Res Soc Sci 67:101510. https://doi.org/10.1016/j.erss.2020.101510

Keahey J (2021) Sustainable development and participatory action research: a systematic review. Syst Pract Action Res 34:291–306. https://doi.org/10.1007/s11213-020-09535-8

Khan A, Becker K (2020) US infrastructure: challenges and directions for the 21st century. Routledge, New York, NY

Kimberlé WC (1989) Demarginalizing the intersection of race and sex: a black feminist critique of antidiscrimination doctrine, feminist theory and antiracist politics. Univ Chic Leg Forum 139–167

Klein JT (2004) Prospects for transdisciplinarity. Futures 36:515–526. https://doi.org/10.1016/j.futures.2003.10.007

Lamont J (2017) Distributive justice. Routledge, New York, NY

Lawrence R, Despres C (2004) Futures of transdisciplinarity. Futures 36:397–405. https://doi.org/10.1016/j.futures.2003.10.005

Lawrence RL, Daniels SE, Stankey GH (1997) Procedural justice and public involvement in natural resource decision making. Soc Nat Resour 10:577–589. https://doi.org/10.1080/08941929709381054

Lee Y-C (2020) Communicating sustainable development: effects of stakeholder-centric perceived sustainability. Corp Soc Responsib Environ Manag 27:1540–1551. https://doi.org/10.1002/csr.1900

Lindgren BM, Lundman B, Graneheim UH (2020) Abstraction and interpretation during the qualitative content analysis process. Int J Nurs Stud 108:103632. https://doi.org/10.1016/j.ijnurstu.2020.103632

Low J (2019) A pragmatic definition of the concept of theoretical saturation. Sociol Focus 52:131–139. https://doi.org/10.1080/00380237.2018.1544514

MacDonald G (2020) Climate, capital, conflict: geographies of success or failure in the twenty-first century. Ann Am Assoc Geogr 110:2011–2031. https://doi.org/10.1080/24694452.2020.1800300

Martens K, Golub A, Robinson G (2012) A justice-theoretic approach to the distribution of transportation benefits: implications for transportation planning practice in the United States. Transp Res Policy Pract 46:684–695. https://doi.org/10.1016/j.tra.2012.01.004

Mertens DM (2009) Transformative research and evaluation. Guilford Press, New York, NY

National Science Foundation (2021) Sustainable regional systems research networks National Science Foundation, Alexandria, Virginia

National Archives and Records Administration (1980) Code of federal regulations: Nondiscrimination in federally-assisted programs of TVA-effectuation of title VI of the civil rights act of. C.F.R, p 18

Pereira RHM, Schwanen T, Banister D (2017) Distributive justice and equity in transportation. Transp Rev 37:170–191. https://doi.org/10.1080/01441647.2016.1257660

Pinson D (2004) Urban planning: an 'undisciplined' discipline? Futures 36:503–513. https://doi.org/10.1016/j.futures.2003.10.008

Prieger JE (2013) The broadband digital divide and the economic benefits of mobile broadband for rural areas. Telecommun Policy 37:483–502. https://doi.org/10.1016/j.telpol.2012.11.003

Ramadier T (2004) Transdisciplinarity and its challenges: the case of urban studies. Futures 36:423–439. https://doi.org/10.1016/j.futures.2003.10.009

Reddick CG, Enriquez R, Harris RJ, Sharma B (2020) Determinants of broadband access and affordability: an analysis of a community survey on the digital divide. Cities 106:102904. https://doi.org/10.1016/j.cities.2020.102904

Saunders B, Sim J, Kingstone T, Baker S, Waterfield J, Bartlam B, Burroughs H, Jinks C (2018) Saturation in qualitative research: exploring its conceptualization and operationalization. Qual Quant 52:1893–1907. https://doi.org/10.1007/s11135-017-0574-8

Simpson LB (2017) As we have always done: Indigenous freedom through radical resistance. University of Minnesota Press, Minneapolis, Minnesota

Singh R, Orazem PF, Song M (2006) Broadband access, telecommuting and the urban-rural digital divide. IOWA State Unversity, United States, US

Sovacool BK, Dworkin MH (2015) Energy justice: conceptual insights and practical applications. Appl Energy 142:435–444. https://doi.org/10.1016/j.apenergy.2015.01.002

Sovacool BK, Newell P, Carley S, Fanzo J (2022) Equity, technological innovation and sustainable behaviour in a low-carbon future. Nat Hum Behav 6(3):326–337

TVA (2023) https://www.tva.com. Accessed 2023

TVA (2024) Regional energy resource council. https://www.tva.com/Community/Regional-Energy-Resource-Council

United Nations (2022) The sustainable development goals report 2022. United Nations, New York, NY

van Bruggen A, Nikolic I, Kwakkel J (2019) Modeling with stakeholders for transformative change. Sustainability 11:825. https://doi.org/10.3390/su11030825

Amanda Fackler is an Assistant Professor in the Department of Social Work at the University of North Dakota. Her research investigates how structural inequalities affect access to resources and sustain disparities among marginalized groups, with special attention to homelessness, housing insecurity, and the impact of the changing environment. Dr. Fackler's work is interdisciplinary, drawing on mixed-methods and community-engaged approaches to address complex issues, inform policy, and develop more equitable and sustainable solutions.

Courtney Cronley is a Professor in the College of Social Work at the University of Tennessee. Her interdisciplinary research explores homelessness, housing precarity, and how the built environment shapes access to opportunity, health, and well-being. A Faculty Fellow at the UT Center for Transportation Research, her mixed-methods, community-based work—funded by the National Science Foundation and the Department of Energy—develops innovative solutions to reduce spatial and environmental barriers to independence and economic mobility.

Janey Camp is a Research Professor in Civil Engineering at the University of Memphis. She is also a licensed civil engineer, a certified GIS Professional, and certified floodplain manager in the state of Tennessee. Dr. Camp's research utilizes tools and models to address today's challenging issues related to infrastructure management and community resilience. Projects involving Dr. Camp's expertise primarily include investigating the impacts of disruptive events on transportation and water infrastructure systems and the communities they serve.

Jonathan Gilligan is Professor of Earth & Environmental Sciences, Professor of Civil & Environmental Engineering, and Professor of Climate & Environmental Studies at Vanderbilt University. Their research focuses on interdisciplinary studies of coupled human and environmental systems, and includes studies of the impacts of environmental change on society and applications of social and behavioral science to designing and informing environmental policy.